2.62 Conics: Composite Transformations

Worked Solutions

1 a)

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (y, -x)$.

$$\frac{x^2}{3} + \frac{y^2}{4} = 1 \mapsto \frac{y^2}{3} + \frac{x^2}{4} = 1 \quad \therefore \quad \frac{x^2}{4} + \frac{y^2}{3} = 1.$$

Enlargement of scale factor 2: $(x, y) \mapsto (\frac{x}{2}, \frac{y}{2})$.

$$\frac{x^2}{4} + \frac{y^2}{3} = 1 \mapsto \frac{(x/2)^2}{4} + \frac{(y/2)^2}{3} = 1 \quad \therefore \quad \frac{x^2}{16} + \frac{y^2}{12} = 1.$$

1 b)

Reflection in the x-axis: $(x, y) \mapsto (x, -y)$.

$$\frac{x^2}{3} + \frac{y^2}{4} = 1 \mapsto \frac{x^2}{3} + \frac{y^2}{4} = 1.$$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (y, -x)$.

$$\frac{x^2}{3} + \frac{y^2}{4} = 1 \mapsto \frac{y^2}{3} + \frac{x^2}{4} = 1 \quad \therefore \quad \frac{x^2}{4} + \frac{y^2}{3} = 1.$$

1 c)

Translation by vector $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$: $(x, y) \mapsto (x - 1, y + 2)$.

$$\frac{x^2}{3} + \frac{y^2}{4} = 1 \mapsto \frac{(x-1)^2}{3} + \frac{(y+2)^2}{4} = 1.$$

Reflection in $y = -x$: $(x, y) \mapsto (-y, -x)$.

$$\frac{(x-1)^2}{3} + \frac{(y+2)^2}{4} = 1 \mapsto \frac{(-y-1)^2}{3} + \frac{(-x+2)^2}{4} = 1 \quad \therefore \quad \frac{(x-2)^2}{4} + \frac{(y+1)^2}{3} = 1.$$

1 d)

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (-y, x)$.

$$\frac{x^2}{3} + \frac{y^2}{4} = 1 \mapsto \frac{y^2}{3} + \frac{x^2}{4} = 1.$$

Enlargement of scale factor $\frac{1}{2}$: $(x, y) \mapsto (2x, 2y)$.

$$\frac{y^2}{3} + \frac{x^2}{4} = 1 \mapsto \frac{(2y)^2}{3} + \frac{(2x)^2}{4} = 1 \quad \therefore \quad x^2 + \frac{4y^2}{3} = 1.$$

1 e)

Stretch parallel to the x-axis, scale factor 3: $(x, y) \mapsto (\frac{x}{3}, y)$.

$$\frac{x^2}{3} + \frac{y^2}{4} = 1 \mapsto \frac{(x/3)^2}{3} + \frac{y^2}{4} = 1 \quad \therefore \quad \frac{x^2}{27} + \frac{y^2}{4} = 1.$$

Translation by vector $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$: $(x, y) \mapsto (x + 1, y - 1)$.

$$\frac{x^2}{27} + \frac{y^2}{4} = 1 \mapsto \frac{(x+1)^2}{27} + \frac{(y-1)^2}{4} = 1.$$

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (-y, x)$.

$$\frac{(x+1)^2}{27} + \frac{(y-1)^2}{4} = 1 \mapsto \frac{(-y+1)^2}{27} + \frac{(x-1)^2}{4} = 1 \quad \therefore \quad \frac{(x-1)^2}{4} + \frac{(y-1)^2}{27} = 1.$$

2 a)

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (-y, x)$.

$$\frac{x^2}{6} - \frac{y^2}{3} = 1 \mapsto \frac{y^2}{6} - \frac{x^2}{3} = 1.$$

Stretch parallel to the y-axis, scale factor 4: $(x, y) \mapsto (x, \frac{y}{4})$.

$$\frac{y^2}{6} - \frac{x^2}{3} = 1 \mapsto \frac{(y/4)^2}{6} - \frac{x^2}{3} = 1 \quad \therefore \quad \frac{y^2}{96} - \frac{x^2}{3} = 1.$$

2 b)

Reflection in the y-axis: $(x, y) \mapsto (-x, y)$.

$$\frac{x^2}{6} - \frac{y^2}{3} = 1 \mapsto \frac{x^2}{6} - \frac{y^2}{3} = 1.$$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (y, -x)$.

$$\frac{x^2}{6} - \frac{y^2}{3} = 1 \mapsto \frac{y^2}{6} - \frac{(-x)^2}{3} = 1 \quad \therefore \quad \frac{y^2}{6} - \frac{x^2}{3} = 1.$$

2 c)

Translation by vector $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$: $(x, y) \mapsto (x + 1, y - 3)$.

$$\frac{x^2}{6} - \frac{y^2}{3} = 1 \mapsto \frac{(x+1)^2}{6} - \frac{(y-3)^2}{3} = 1.$$

Reflection in $y = x$: $(x, y) \mapsto (y, x)$.

$$\frac{(x+1)^2}{6} - \frac{(y-3)^2}{3} = 1 \mapsto \frac{(y+1)^2}{6} - \frac{(x-3)^2}{3} = 1.$$

2 d)

Stretch parallel to the x-axis, scale factor $\frac{1}{3}$: $(x, y) \mapsto (3x, y)$.

$$\frac{x^2}{6} - \frac{y^2}{3} = 1 \mapsto \frac{(3x)^2}{6} - \frac{y^2}{3} = 1.$$

Reflection in $y = -x$: $(x, y) \mapsto (-y, -x)$.

$$\frac{3x^2}{2} - \frac{y^2}{3} = 1 \mapsto \frac{3y^2}{2} - \frac{x^2}{3} = 1.$$

2 e)

Enlargement of scale factor 2: $(x, y) \mapsto (\frac{x}{2}, \frac{y}{2})$.

$$\frac{x^2}{6} - \frac{y^2}{3} = 1 \mapsto \frac{(x/2)^2}{6} - \frac{(y/2)^2}{3} = 1 \quad \therefore \quad \frac{x^2}{24} - \frac{y^2}{12} = 1.$$

Translation by vector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$: $(x, y) \mapsto (x - 1, y + 1)$.

$$\frac{x^2}{24} - \frac{y^2}{12} = 1 \mapsto \frac{(x-1)^2}{24} - \frac{(y+1)^2}{12} = 1.$$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (y, -x)$.

$$\frac{(x-1)^2}{24} - \frac{(y+1)^2}{12} = 1 \mapsto \frac{(y-1)^2}{24} - \frac{(-x+1)^2}{12} = 1 \quad \therefore \quad \frac{(y-1)^2}{24} - \frac{(x-1)^2}{12} = 1.$$

3 a)

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (-y, x)$.

$$xy = 16 \mapsto -xy = 16 \quad \Rightarrow \quad xy = -16.$$

Enlargement of scale factor 3: $(x, y) \mapsto (\frac{x}{3}, \frac{y}{3})$.

$$xy = -16 \mapsto \left(\frac{x}{3}\right)\left(\frac{y}{3}\right) = -16 \quad \therefore \quad xy = -144.$$

3 b)

Reflection in the x-axis: $(x, y) \mapsto (x, -y)$.

$$xy = 16 \mapsto -xy = 16 \quad \Rightarrow \quad xy = -16.$$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (y, -x)$.

$$xy = -16 \mapsto xy = 16.$$

3 c)

Translation by vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$: $(x, y) \mapsto (x - 1, y - 2)$.

$$xy = 16 \mapsto (x-1)(y-2) = 16.$$

Reflection in $y = -x$: $(x, y) \mapsto (-y, -x)$.

$$(x-1)(y-2) = 16 \mapsto (-y-1)(-x-2) = 16 \quad \therefore \quad (y+1)(x+2) = 16.$$

3 d)

Stretch parallel to the x-axis, scale factor 2: $(x, y) \mapsto (\frac{x}{2}, y)$.

$$xy = 16 \mapsto \left(\frac{x}{2}\right) y = 16 \quad \Rightarrow \quad xy = 32.$$

Translation by vector $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$: $(x, y) \mapsto (x + 2, y - 4)$.

$$xy = 32 \mapsto (x + 2)(y - 4) = 32.$$

3 e)

Enlargement of scale factor 3: $(x, y) \mapsto (\frac{x}{2}, \frac{y}{2})$.

$$xy = 16 \mapsto \left(\frac{x}{2}\right)\left(\frac{y}{2}\right) = 16 \quad \Rightarrow \quad xy = 64.$$

Translation by vector $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$: $(x, y) \mapsto (x + 1, y - 4)$.

$$xy = 64 \mapsto (x + 1)(y - 4) = 64.$$

Rotate by $\frac{3\pi}{2}$ radians anti-clockwise: $(x, y) \mapsto (-y, x)$.

$$(x + 1)(y - 4) = 64 \mapsto (-y + 1)(x - 4) = 64 \quad \therefore \quad (y - 1)(x - 4) = -64.$$

4 a)

Stretch parallel to the y-axis, scale factor 2: $(x, y) \mapsto (x, \frac{y}{2})$.

$$\frac{x^2}{5} + \frac{y^2}{12} = 1 \mapsto \frac{x^2}{5} + \frac{(y/2)^2}{12} = 1 \quad \Rightarrow \quad \frac{x^2}{5} + \frac{y^2}{48} = 1.$$

Translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$: $(x, y) \mapsto (x - 3, y + 4)$.

$$\frac{x^2}{5} + \frac{y^2}{48} = 1 \mapsto \frac{(x - 3)^2}{5} + \frac{(y + 4)^2}{48} = 1.$$

Rotate by π radians anti-clockwise: $(x, y) \mapsto (-x, -y)$.

$$\frac{(x - 3)^2}{5} + \frac{(y + 4)^2}{48} = 1 \mapsto \frac{(-x - 3)^2}{5} + \frac{(-y + 4)^2}{48} = 1 \quad \therefore \quad \frac{(x + 3)^2}{5} + \frac{(y - 4)^2}{48} = 1.$$

4 b)

Stretch parallel to the y-axis, scale factor 2: $(x, y) \mapsto (x, \frac{y}{2})$.

$$\frac{x^2}{4} - \frac{y^2}{6} = 1 \mapsto \frac{x^2}{4} - \frac{(y/2)^2}{6} = 1 \quad \Rightarrow \quad \frac{x^2}{4} - \frac{y^2}{24} = 1.$$

Translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$: $(x, y) \mapsto (x - 3, y + 4)$.

$$\frac{x^2}{4} - \frac{y^2}{24} = 1 \mapsto \frac{(x - 3)^2}{4} - \frac{(y + 4)^2}{24} = 1.$$

Rotate by π radians anti-clockwise: $(x, y) \mapsto (-x, -y)$.

$$\frac{(x - 3)^2}{4} - \frac{(y + 4)^2}{24} = 1 \mapsto \frac{(-x - 3)^2}{4} - \frac{(-y + 4)^2}{24} = 1 \quad \therefore \quad \frac{(x + 3)^2}{4} - \frac{(y - 4)^2}{24} = 1.$$

4 c)

Stretch parallel to the y-axis, scale factor 2: $(x, y) \mapsto (x, \frac{y}{2})$.

$$xy = 11 \mapsto x\left(\frac{y}{2}\right) = 11 \quad \Rightarrow \quad xy = 22.$$

Translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$: $(x, y) \mapsto (x - 3, y + 4)$.

$$xy = 22 \mapsto (x - 3)(y + 4) = 22.$$

Rotate by π radians anti-clockwise: $(x, y) \mapsto (-x, -y)$.

$$(x - 3)(y + 4) = 22 \mapsto (-x - 3)(-y + 4) = 22 \quad \therefore \quad (x + 3)(y - 4) = 22.$$

4 d)

Stretch parallel to the y-axis, scale factor 2: $(x, y) \mapsto (x, \frac{y}{2})$.

$$x^2 - y^2 = 9 \mapsto x^2 - (y/2)^2 = 9 \quad \Rightarrow \quad x^2 - \frac{y^2}{4} = 9.$$

Translation by vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$: $(x, y) \mapsto (x - 3, y + 4)$.

$$x^2 - \frac{y^2}{4} = 9 \mapsto (x - 3)^2 - \frac{(y + 4)^2}{4} = 9.$$

Rotate by π radians anti-clockwise: $(x, y) \mapsto (-x, -y)$.

$$(x - 3)^2 - \frac{(y + 4)^2}{4} = 9 \mapsto (-x - 3)^2 - \frac{(-y + 4)^2}{4} = 9 \quad \therefore \quad (x + 3)^2 - \frac{(y - 4)^2}{4} = 9.$$

5 a)

$$x^2 + 9y^2 = 1 \mapsto \frac{x^2}{36} + \frac{y^2}{4} = 1 \mapsto \frac{(x - 2)^2}{36} + \frac{y^2}{4} = 1.$$

Enlargement of scale factor 6 about $(0, 0)$ followed by a translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Or

$$x^2 + 9y^2 = 1 \mapsto \left(x - \frac{1}{3}\right)^2 + 9y^2 = 1 \mapsto \frac{(x - 2)^2}{36} + \frac{y^2}{4} = 1.$$

Translation by vector $\begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix}$ followed by an enlargement of scale factor 6 about $(0, 0)$.

5 b)

$$\frac{x^2}{9} - \frac{y^2}{4} = 1 \mapsto \frac{x^2}{9} - \frac{y^2}{16} = 1 \mapsto \frac{(x - 3)^2}{9} - \frac{y^2}{16} = 1.$$

Stretch parallel to the y-axis, scale factor 2, and translation by vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, in either order.

5 c)

$$xy = 1 \mapsto (x-2)y = 1 \mapsto (x-2)y = 3.$$

Translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ followed by a stretch parallel to the y-axis, scale factor 3, in either order.

Or

$$xy = 1 \mapsto \left(\frac{x}{3}\right) y = 1 \mapsto (x-2)y = 3.$$

Stretch parallel to the x-axis, scale factor 3, followed by a translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Or

$$xy = 1 \mapsto xy = 3 \mapsto (x-2)y = 3.$$

Enlargement of scale factor $\sqrt{3}$ about $(0, 0)$ followed by a translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

5 d)

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \mapsto \frac{x^2}{16} + \frac{y^2}{36} = 1 \mapsto \frac{(x-1)^2}{16} + \frac{(y+3)^2}{36} = 1.$$

Stretch parallel to the y-axis, scale factor 2, followed by a translation by vector $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$. Or

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \mapsto \frac{(x-1)^2}{16} + \frac{(y+\frac{3}{2})^2}{9} = 1 \mapsto \frac{(x-1)^2}{16} + \frac{(y+3)^2}{36} = 1.$$

Translation by vector $\begin{pmatrix} 1 \\ -\frac{3}{2} \end{pmatrix}$ followed by a stretch parallel to the y-axis, scale factor 2.

5 e)

$$\frac{x^2}{4} - \frac{y^2}{9} = 1 \mapsto \frac{x^2}{8} - \frac{y^2}{18} = 1 \mapsto \frac{(x-2)^2}{8} - \frac{y^2}{18} = 1 \Rightarrow \frac{(2-x)^2}{8} - \frac{y^2}{18} = 1.$$

Enlargement of scale factor $\sqrt{2}$ about $(0, 0)$ followed by a translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Or

$$\frac{x^2}{4} - \frac{y^2}{9} = 1 \mapsto \frac{(x-\sqrt{2})^2}{4} - \frac{y^2}{9} = 1 \mapsto \frac{(x-2)^2}{8} - \frac{y^2}{18} = 1 \Rightarrow \frac{(2-x)^2}{8} - \frac{y^2}{18} = 1.$$

Translation by vector $\begin{pmatrix} \sqrt{2} \\ 0 \end{pmatrix}$ followed by an enlargement of scale factor $\sqrt{2}$ about $(0, 0)$.

5 f)

$$\frac{x^2}{25} + \frac{y^2}{4} = 1 \mapsto \frac{x^2}{4} + \frac{y^2}{25} = 1 \mapsto \frac{(x+2)^2}{4} + \frac{(y-1)^2}{25} = 1.$$

Rotate by $\frac{\pi}{2}$ radians anti-clockwise, or rotate by $\frac{3\pi}{2}$ radians anti-clockwise, or reflect in $y = x$, or reflect in $y = -x$, and translate by vector $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$, in either order.

5 g)

$$xy = 1 \mapsto xy = 4 \mapsto (x+1)(y-2) = 4.$$

Enlargement of scale factor 2 about $(0,0)$ followed by a translation by vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$. Or

$$xy = 1 \mapsto \left(\frac{x}{4}\right) y = 1 \mapsto (x+1)(y-2) = 4.$$

Stretch parallel to the x-axis, scale factor 4, followed by a translation by vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$. Or

$$xy = 1 \mapsto x\left(\frac{y}{4}\right) = 1 \mapsto (x+1)(y-2) = 4.$$

Stretch parallel to the y-axis, scale factor 4, followed by a translation by vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$. Or

$$xy = 1 \mapsto \left(x + \frac{1}{2}\right)(y-1) = 1 \mapsto \left(\frac{1}{2}x + \frac{1}{2}\right)\left(\frac{1}{2}y - 1\right) = 1 \Rightarrow (x+1)(y-2) = 4.$$

Translation by vector $\begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix}$ followed by an enlargement of scale factor 2 about $(0,0)$.

5 h)

$$9(y-1)^2 - 4x^2 = 144 \Rightarrow \frac{9(y-1)^2}{144} - \frac{4x^2}{144} = 1 \Rightarrow \frac{(y-1)^2}{16} - \frac{x^2}{36} = 1.$$

$$\frac{y^2}{16} - \frac{x^2}{9} = 1 \mapsto \frac{y^2}{16} - \frac{x^2}{36} = 1 \mapsto \frac{(y-1)^2}{16} - \frac{x^2}{36} = 1.$$

Stretch parallel to the x-axis, scale factor 2, and translate by vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, in either order.

5 i)

$$x^2 + 5(y^2 - 4y) = -5 \Rightarrow x^2 + 5(y-2)^2 - 20 = -5 \Rightarrow x^2 + 5(y-2)^2 = 15 \Rightarrow \frac{x^2}{15} + \frac{(y-2)^2}{3} = 1.$$

$$\frac{x^2}{5} + \frac{y^2}{3} = 1 \mapsto \frac{x^2}{5} + \frac{(y-2)^2}{3} = 1 \mapsto \frac{x^2}{15} + \frac{(y-2)^2}{3} = 1.$$

Stretch parallel to the x-axis, scale factor $\sqrt{3}$, followed by a translation by vector $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$, in either order.

5 j)

$$\begin{aligned} x^2 = 5(y^2 + 2x) &\Rightarrow x^2 - 10x - 5y^2 = 0 \\ &\Rightarrow (x-5)^2 - 25 - 5y^2 = 0 \\ &\Rightarrow (x-5)^2 - 5y^2 = 25 \\ &\Rightarrow \frac{(x-5)^2}{25} - \frac{y^2}{5} = 1. \end{aligned}$$

$$x^2 - \frac{y^2}{5} = 1 \mapsto \frac{x^2}{25} - \frac{y^2}{5} = 1 \mapsto \frac{(x-5)^2}{25} - \frac{y^2}{5} = 1.$$

Stretch parallel to the x-axis, scale factor 5, followed by a translation by vector $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$. Or

$$x^2 - \frac{y^2}{5} = 1 \mapsto (x-1)^2 - \frac{y^2}{5} = 1 \mapsto \frac{(x-5)^2}{25} - \frac{y^2}{5} = 1.$$

Translation by vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ followed by a stretch parallel to the x-axis, scale factor 5.

6 a)

$$2(x-1)^2 + (y+3)^2 = 16 \Rightarrow \frac{(x-1)^2}{8} + \frac{(y+3)^2}{16} = 1.$$

$$\frac{x^2}{2} + \frac{y^2}{4} = 1 \mapsto \frac{x^2}{8} + \frac{y^2}{16} = 1 \mapsto \frac{(x-1)^2}{8} + \frac{(y+3)^2}{16} = 1.$$

Enlargement of scale factor 2 about $(0, 0)$ followed by a translation by vector $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$. Or

$$\frac{x^2}{2} + \frac{y^2}{4} = 1 \mapsto \frac{(x-\frac{1}{2})^2}{2} + \frac{(y+\frac{3}{2})^2}{4} = 1 \mapsto \frac{(x-1)^2}{8} + \frac{(y+3)^2}{16} = 1.$$

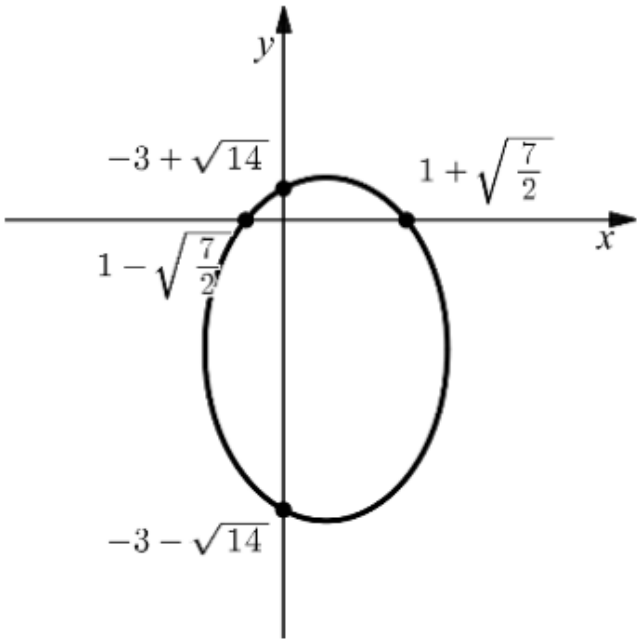
Translation by vector $\begin{pmatrix} \frac{1}{2} \\ -\frac{3}{2} \end{pmatrix}$ followed by an enlargement of scale factor 2 about $(0, 0)$.

6 b)

When $x = 0$, $2(0-1)^2 + (y+3)^2 = 16 \Rightarrow y = -3 \pm \sqrt{14}$. When $y = 0$, $2(x-1)^2 + 3^2 = 16 \Rightarrow x = 1 \pm \sqrt{\frac{7}{2}}$.

6 c)

Sketch of the curve C_2.



7 a)

Hyperbola.

7 b)

Asymptotes are $y = \frac{4}{3}x$ and $y = -\frac{4}{3}x$.

7 c)

Translation by vector $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$: $(x, y) \mapsto (x - 4, y - 1)$.

$$y = \pm\frac{4}{3}x \mapsto y - 1 = \pm\frac{4}{3}(x - 4).$$

Reflection in the line $y = x$: $(x, y) \mapsto (y, x)$.

$$y - 1 = \pm\frac{4}{3}(x - 4) \mapsto x - 1 = \pm\frac{4}{3}(y - 4).$$

So

$$x - 1 = \frac{4}{3}(y - 4) \Rightarrow y = \frac{3}{4}x + \frac{13}{4}, \qquad x - 1 = -\frac{4}{3}(y - 4) \Rightarrow y = -\frac{3}{4}x + \frac{19}{4}.$$

$$\therefore y = \frac{3}{4}x + \frac{13}{4} \quad \text{or} \quad y = -\frac{3}{4}x + \frac{19}{4}.$$

8 a)

Rotation by $\frac{\pi}{2}$ anti-clockwise followed by an enlargement about the origin, scale factor 2 (although the order does not matter).

8 b)

$$M = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}.$$

$$\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2y \\ 2x \end{pmatrix}.$$

So $X = -2y$ and $Y = 2x$, giving $x = \frac{Y}{2}$ and $y = -\frac{X}{2}$.

$$x^2 + 4y^2 = 1 \mapsto \left(\frac{Y}{2}\right)^2 + 4\left(-\frac{X}{2}\right)^2 = 1 \Rightarrow \frac{Y^2}{4} + X^2 = 1.$$

$$\therefore x^2 + \frac{y^2}{4} = 1.$$

9 a)

Enlargement about the origin, scale factor 4, followed by a reflection across the x-axis.

9 b)

$$M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & -4 \end{pmatrix}.$$

$$\begin{pmatrix} 4 & 0 \\ 0 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4x \\ -4y \end{pmatrix}.$$

So $X = 4x$ and $Y = -4y$, giving $x = \frac{X}{4}$ and $y = -\frac{Y}{4}$.

$$x^2 - y^2 = 4 \mapsto \left(\frac{X}{4}\right)^2 - \left(-\frac{Y}{4}\right)^2 = 4 \Rightarrow \frac{X^2}{16} - \frac{Y^2}{16} = 4.$$

$$\therefore x^2 - y^2 = 64.$$

10 a)

$$\frac{\sqrt{2}}{\sqrt{8}} = \frac{1}{2} \quad \therefore \quad y = \frac{1}{2}x \text{ or } y = -\frac{1}{2}x.$$

$y = \frac{1}{2}x$ has direction vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$:

$$\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}, \quad \text{so } y = -3x.$$

$y = -\frac{1}{2}x$ has direction vector $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$:

$$\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \quad \text{so } y = x.$$

$$\therefore y = -3x \quad \text{and} \quad y = x.$$

10 b)

$$\frac{\sqrt{2}}{\sqrt{8}} = \frac{1}{2} \quad \therefore \quad y = \frac{1}{2}x \text{ or } y = -\frac{1}{2}x.$$

$y = \frac{1}{2}x$ has direction vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$:

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}, \quad \text{so } y = \frac{2}{5}x.$$

$y = -\frac{1}{2}x$ has direction vector $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$:

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \end{pmatrix}, \quad \text{so } y = \frac{2}{3}x.$$

$$\therefore y = \frac{2}{5}x \quad \text{and} \quad y = \frac{2}{3}x.$$

10 c)

$$\frac{\sqrt{2}}{\sqrt{8}} = \frac{1}{2} \quad \therefore \quad y = \frac{1}{2}x \text{ or } y = -\frac{1}{2}x.$$

$y = \frac{1}{2}x$ has direction vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 3 & -1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{5}{\sqrt{2}} \\ \frac{7}{\sqrt{2}} \end{pmatrix}, \quad \text{so } y = \frac{7}{5}x.$$

$y = -\frac{1}{2}x$ has direction vector $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 3 & -1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{7}{\sqrt{2}} \\ -\frac{5}{\sqrt{2}} \end{pmatrix}, \quad \text{so } y = \frac{5}{7}x.$$

$$\therefore y = \frac{7}{5}x \quad \text{and} \quad y = \frac{5}{7}x.$$

10 d)

$$\frac{\sqrt{2}}{\sqrt{8}} = \frac{1}{2} \quad \therefore \quad y = \frac{1}{2}x \text{ or } y = -\frac{1}{2}x.$$

$y = \frac{1}{2}x$ has direction vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$:

$$\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}, \quad \text{so } y = -\frac{1}{4}x.$$

$y = -\frac{1}{2}x$ has direction vector $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$:

$$\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \text{so } x = 0.$$

$$\therefore y = -\frac{1}{4}x \quad \text{and} \quad x = 0.$$

11 a)

$$M = \begin{pmatrix} 0 & -\frac{2}{3} \\ \frac{1}{3} & 0 \end{pmatrix}, \qquad \det M = \frac{2}{9}.$$

11 b)

$$\begin{pmatrix} 0 & -\frac{2}{3} \\ \frac{1}{3} & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{2}{3}y \\ \frac{1}{3}x \end{pmatrix}.$$

So $X = -\frac{2}{3}y$ and $Y = \frac{1}{3}x$, giving $x = 3Y$ and $y = -\frac{3X}{2}$.

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \mapsto \frac{(3Y)^2}{9} + \frac{(-3X/2)^2}{4} = 1 \Rightarrow Y^2 + \frac{9X^2}{16} = 1.$$

$$\therefore \frac{9x^2}{16} + y^2 = 1.$$

11 c)

$$6\pi \times \frac{2}{9} = \frac{4}{3}\pi \text{ units}^2.$$

12 a)

$\det M = 3$.

12 b)

$$\begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ -x+2y \end{pmatrix}.$$

So $X = x + y$ and $Y = -x + 2y$.

$$X + Y = 3y \Rightarrow y = \frac{1}{3}(X + Y).$$

$$Y - X = y - 2x = \frac{1}{3}X + \frac{1}{3}Y - 2x \Rightarrow 2x = \frac{4}{3}X - \frac{2}{3}Y \Rightarrow x = \frac{2}{3}X - \frac{1}{3}Y.$$

So $x = \frac{2}{3}X - \frac{1}{3}Y$ and $y = \frac{1}{3}X + \frac{1}{3}Y$.

$$x^2 + y^2 = 9 \mapsto \left(\frac{2}{3}X - \frac{1}{3}Y\right)^2 + \left(\frac{1}{3}X + \frac{1}{3}Y\right)^2 = 9.$$

$$\Rightarrow \frac{4}{9}X^2 - \frac{4}{9}XY + \frac{1}{9}Y^2 + \frac{1}{9}X^2 + \frac{2}{9}XY + \frac{1}{9}Y^2 = 9 \Rightarrow \frac{5}{9}X^2 - \frac{2}{9}XY + \frac{2}{9}Y^2 = 9.$$

$$\therefore 5x^2 - 2xy + 2y^2 = 81.$$

12 c)

The original curve has area 9π.

$$3 \times 9\pi = 27\pi \text{ units}^2.$$

13 a)

$$\frac{1}{2}\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2}x - \frac{1}{2}y \\ \frac{1}{2}y \end{pmatrix}.$$

So $X = \frac{1}{2}x - \frac{1}{2}y$ and $Y = \frac{1}{2}y$. Thus $X + Y = \frac{1}{2}x$, so $x = 2X + 2Y$ and $y = 2Y$.

$$4x^2 - 8xy + 8y^2 = 1 \mapsto 4(2X + 2Y)^2 - 8(2X + 2Y)(2Y) + 8(2Y)^2 = 1.$$

$$\Rightarrow 16X^2 + 32XY + 16Y^2 - 32XY - 32Y^2 + 32Y^2 = 1 \Rightarrow 16X^2 + 16Y^2 = 1.$$

$$\therefore x^2 + y^2 = \frac{1}{16}.$$

13 b)

It is a circle, so C' has area $\frac{\pi}{16}$ units^2.

13 c)

$$\det \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{4}.$$

So $\frac{1}{4} \times$ area of $C = \frac{\pi}{16}$.

$$\therefore \text{area of } C = \frac{\pi}{4} \text{ units}^2.$$

14 a)

$y = \frac{1}{2}x$ and $y = -\frac{1}{2}x$.

14 b)

$$x^2 + 8x - y^2 = (x+4)^2 - 16 - y^2.$$

So $(x+4)^2 - y^2 = 16$, hence $\frac{(x+4)^2}{16} - \frac{y^2}{16} = 1$.

$$\frac{x^2}{16} - \frac{y^2}{4} = 1 \mapsto \frac{(x+4)^2}{16} - \frac{y^2}{4} = 1 \mapsto \frac{(x+4)^2}{16} - \frac{y^2}{16} = 1.$$

Translation by vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$ and a stretch parallel to the y-axis, scale factor 2, in either order.

14 c)

$$y = \frac{1}{2}x \mapsto \frac{y}{2} = \frac{1}{2}(x+4), \qquad y = -\frac{1}{2}x \mapsto \frac{y}{2} = -\frac{1}{2}(x+4).$$

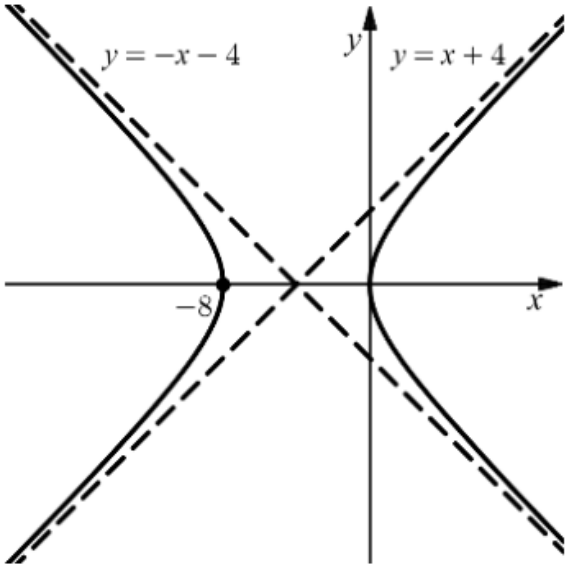
$$\therefore y = x + 4 \quad \text{and} \quad y = -x - 4.$$

14 d)

When $x = 0$, $y = 0$; and when $y = 0$, $x = 0$ or $x = -8$.

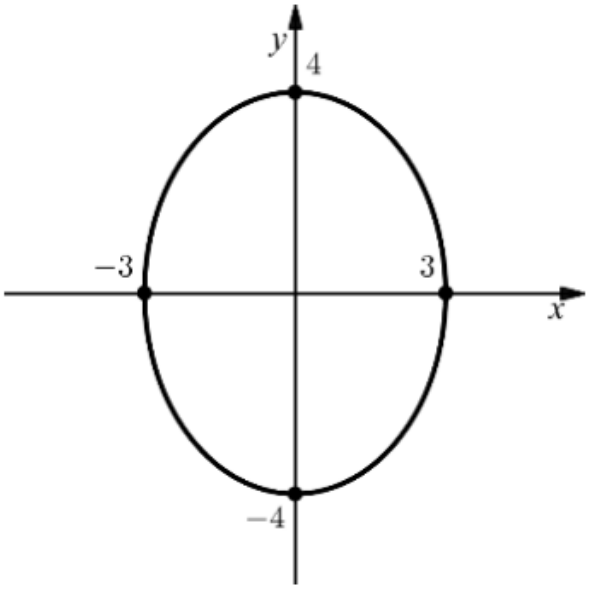
14 e)

Sketch of C_2.



15 a)

Sketch of C.



15 b)

$$\frac{x^2}{9} + \frac{y^2}{16} = 1.$$

Stretch parallel to the y-axis, scale factor $\frac{1}{4}$:

$$\frac{x^2}{9} + \frac{y^2}{16} = 1 \mapsto \frac{x^2}{9} + \frac{(4y)^2}{16} = 1 \Rightarrow \frac{x^2}{9} + y^2 = 1.$$

Stretch parallel to the x-axis, scale factor $\frac{1}{3}$:

$$\frac{x^2}{9} + y^2 = 1 \mapsto \frac{(3x)^2}{9} + y^2 = 1 \Rightarrow x^2 + y^2 = 1.$$

$$\therefore \text{stretch parallel to the } y\text{-axis, scale factor } \frac{1}{4},$$
$$\text{and stretch parallel to the } x\text{-axis, scale factor } \frac{1}{3}, \text{ in either order.}$$

15 c)

$$\pi \times 1^2 \times 4 \times 3 = 12\pi \text{ units}^2.$$

15 d)

$$\frac{(x-p)^2}{9} + \frac{(y-q)^2}{16} = 1.$$

Touching the x-axis: $q = 4$. Touching the y-axis: $p = 3$.

$$\therefore p = 3 \text{ and } q = 4.$$

15 e)

$$\left(\frac{x}{2}\right)^2 /9 + \frac{y^2}{16} = 1 \Rightarrow \frac{x^2}{36} + \frac{y^2}{16} = 1.$$
$$\therefore 4x^2 + 9y^2 = 144.$$

16 a)

When $x = 0$, $y = \pm 1$; when $y = 0$, $x = \pm 4$.

$$\therefore (0,1), (0,-1), (4,0), (-4,0).$$

16 b)

$$6(x^2 + 4x) + y^2 + 2y = -19$$
$$6(x+2)^2 - 24 + (y+1)^2 - 1 = -19$$
$$6(x+2)^2 + (y+1)^2 = 6$$
$$(x+2)^2 + \frac{(y+1)^2}{6} = 1.$$

$$\frac{x^2}{16} + y^2 = 1 \mapsto x^2 + y^2 = 1 \mapsto x^2 + \frac{y^2}{6} = 1 \mapsto (x+2)^2 + \frac{(y+1)^2}{6} = 1.$$

Stretch parallel to the x-axis, scale factor $\frac{1}{4}$, and stretch parallel to the y-axis, scale factor $\sqrt{6}$, in either order, followed by a translation by vector $\begin{pmatrix} -2 \\ -1 \end{pmatrix}$.

16 c)

When $x = 0$, y has no real solutions; when $y = 0$, $x = -2 \pm \sqrt{\frac{5}{6}}$.

$$\therefore \left(-2 + \sqrt{\frac{5}{6}}, 0\right), \quad \left(-2 - \sqrt{\frac{5}{6}}, 0\right).$$

16 d)

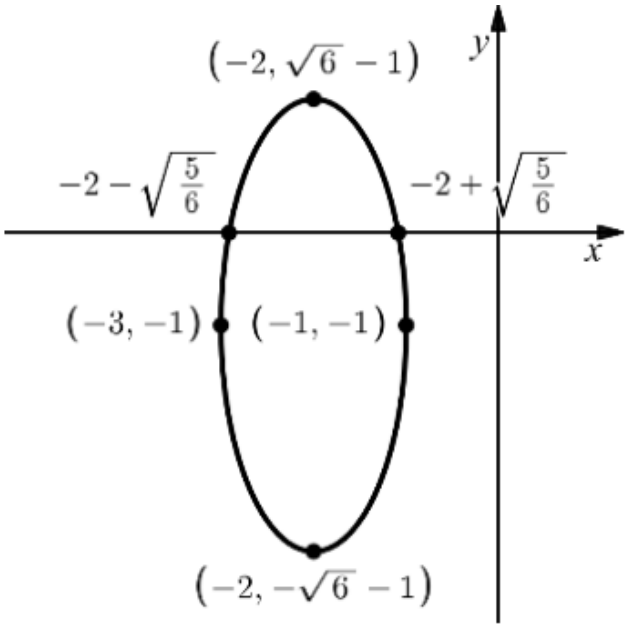
Maximum and minimum values occur on $\frac{x^2}{16} + y^2 = 1$ when $x = \pm 4$ and $y = \pm 1$.

$$\begin{aligned} x = 4 \mapsto x = 1 \mapsto x = 1 \mapsto x = -1, \\ x = -4 \mapsto x = -1 \mapsto x = -1 \mapsto x = -3, \\ y = 1 \mapsto y = 1 \mapsto y = \sqrt{6} \mapsto y = \sqrt{6} - 1, \\ y = -1 \mapsto y = -1 \mapsto y = -\sqrt{6} \mapsto y = -\sqrt{6} - 1. \end{aligned}$$

So $x_{\max} = -1$, $x_{\min} = -3$, $y_{\max} = \sqrt{6} - 1$, $y_{\min} = -\sqrt{6} - 1$.

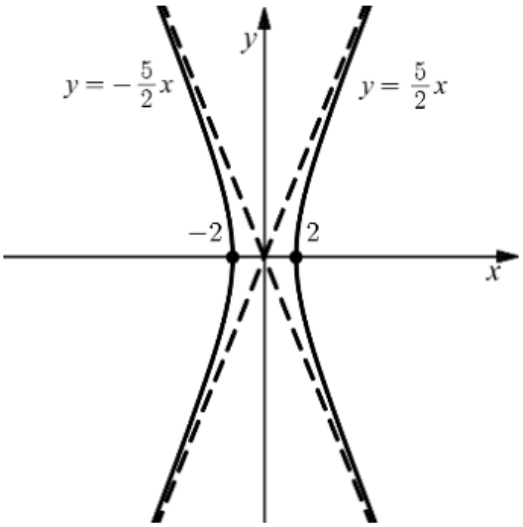
16 e)

When $x = -1$, $y = -1$; when $x = -3$, $y = -1$. When $y = \sqrt{6} - 1$, $x = -2$; when $y = -\sqrt{6} - 1$, $x = -2$.



17 a)

When $x = 0$, $y = \pm 2$.



17 b)

Translation by vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$: $(x, y) \mapsto (x - k, y)$.

$$\frac{x^2}{4} - \frac{y^2}{25} = 1 \mapsto \frac{(x-k)^2}{4} - \frac{y^2}{25} = 1.$$

Reflection in $y = -x$: $(x, y) \mapsto (-y, -x)$.

$$\frac{(x-k)^2}{4} - \frac{y^2}{25} = 1 \mapsto \frac{(-y-k)^2}{4} - \frac{x^2}{25} = 1.$$

This passes through the origin, $(0, 0)$:

$$\frac{(0-k)^2}{4} - \frac{0}{25} = 1 \Rightarrow k^2 = 4.$$

Since k is positive, $k = 2$.

17 c)

Translation by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$: $(x, y) \mapsto (x - 2, y)$.

$$y = \pm\frac{5}{2}x \mapsto y = \pm\frac{5}{2}(x-2).$$

Reflection in $y = -x$: $(x, y) \mapsto (-y, -x)$.

$$y = \pm\frac{5}{2}(x-2) \mapsto -x = \pm\frac{5}{2}(-y-2).$$

So

$$-x = \frac{5}{2}(-y-2) \Rightarrow y = \frac{2}{5}x - 2, \qquad -x = -\frac{5}{2}(-y-2) \Rightarrow y = -\frac{2}{5}x - 2.$$

18 a)

Translation by vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$: $(x, y) \mapsto (x-2, y+1)$.

$$\frac{x^2}{4} - \frac{y^2}{3} = 1 \mapsto \frac{(x-2)^2}{4} - \frac{(y+1)^2}{3} = 1.$$

Rotation anti-clockwise by $\frac{\pi}{2}$: $(x, y) \mapsto (y, -x)$.

$$\frac{(x-2)^2}{4} - \frac{(y+1)^2}{3} = 1 \mapsto \frac{(y-2)^2}{4} - \frac{(-x+1)^2}{3} = 1.$$

Stretch in the x-direction by factor 3: $(x, y) \mapsto (\frac{x}{3}, y)$.

$$\frac{(y-2)^2}{4} - \frac{(-x+1)^2}{3} = 1 \mapsto \frac{(y-2)^2}{4} - \frac{(-x/3+1)^2}{3} = 1 \Rightarrow \frac{(y-2)^2}{4} - \frac{(x-3)^2}{27} = 1.$$

Thus

$$27(y-2)^2 - 4(x-3)^2 = 108 \Rightarrow 27(y^2 - 4y + 4) - 4(x^2 - 6x + 9) = 108$$
$$\Rightarrow 27y^2 - 108y + 108 - 4x^2 + 24x - 36 = 108 \Rightarrow 4x^2 - 27y^2 - 24x + 108y + 36 = 0.$$

18 b)

For $\frac{x^2}{4} - \frac{y^2}{3} = 1$, the points $(-2, 0)$ and $(2, 0)$ are on the hyperbola. Under the transformations:

$$(-2, 0) \mapsto (0, -1) \mapsto (1, 0) \mapsto (3, 0), \qquad (2, 0) \mapsto (4, -1) \mapsto (1, 4) \mapsto (3, 4).$$

So $k < 0$ or $k > 4$.

$$\therefore \{k : k \in \mathbb{R},\ k < 0\} \cup \{k : k \in \mathbb{R},\ k > 4\}.$$

Alternatively, when $y = k$:

$$4x^2 - 27k^2 + 108k - 24x + 36 = 0 \Rightarrow 4x^2 - 24x + 36 + 108k - 27k^2 = 0.$$

Two points of intersection require $b^2 - 4ac > 0$:

$$(-24)^2 - 4(4)(36 + 108k - 27k^2) > 0 \Rightarrow 432k^2 - 1728k > 0 \Rightarrow 432k(k-4) > 0.$$

So $k < 0$ or $k > 4$.

$$\therefore \{k : k \in \mathbb{R},\ k < 0\} \cup \{k : k \in \mathbb{R},\ k > 4\}.$$

19 a)

$$4x - 3y = 11 \Rightarrow y = \frac{4}{3}x - \frac{11}{3}.$$
$$\frac{x^2}{9} - \frac{y^2}{16} = 1 \Rightarrow \frac{x^2}{9} - \frac{(\frac{4}{3}x - \frac{11}{3})^2}{16} = 1 \Rightarrow \frac{x^2}{9} - \frac{(4x-11)^2}{144} = 1.$$
$$16x^2 - (4x-11)^2 = 144 \Rightarrow 16x^2 - 16x^2 + 88x - 121 = 144 \Rightarrow x = \frac{265}{88}.$$

There is one solution, and so only one point of intersection.

19 b)

The asymptote $y = \frac{4}{3}x$ is translated:

$$y + 1 = \frac{4}{3}(x - a) \Rightarrow 3y + 3 = 4x - 4a \Rightarrow 4a + 3 = 4x - 3y.$$

Since $4x - 3y = 11$ is now an asymptote, $4a + 3 = 11$.

$$\therefore a = 2.$$

20 a)

Consider the points $(5, 0)$, $(0, 2)$, $(-5, 0)$ and $(0, -2)$ and apply the reflection across $y = x$, translation by vector $\begin{pmatrix} a \\ b \end{pmatrix}$, and enlargement of scale factor 2. $(-5, 0)$ is transformed to the x-axis, and $(0, -2)$ is transformed to the y-axis:

$$(-5, 0) \mapsto (0, -5) \mapsto (a, -5 + b) \mapsto (2a, -10 + 2b).$$

$$-10 + 2b = 0 \Rightarrow b = 5.$$

$$(0, -2) \mapsto (-2, 0) \mapsto (a - 2, b) \mapsto (2a - 4, 2b).$$

$$2a - 4 = 0 \Rightarrow a = 2.$$

$$\therefore a = 2 \text{ and } b = 5.$$

20 b)

$$\frac{x^2}{25} + \frac{y^2}{4} = 1 \mapsto \frac{x^2}{4} + \frac{y^2}{25} = 1 \mapsto \frac{(x-2)^2}{4} + \frac{(y-5)^2}{25} = 1.$$

After the enlargement of scale factor 2:

$$\frac{(x/2 - 2)^2}{4} + \frac{(y/2 - 5)^2}{25} = 1 \Rightarrow \frac{(x-4)^2}{16} + \frac{(y-10)^2}{100} = 1.$$

21 a)

Ellipse.

21 b)

The translation must have negative values for both a and b so that the curve is translated to the left and downwards. After reflection in the y-axis, it touches C_1 just once; the maximum point of C_2 must be $(2, -3)$. This point is reflected from $(-2, -3)$. Therefore the maximum of C_1 is translated from $(2, 3)$ to $(-2, -3)$, which is translation $\begin{pmatrix} -4 \\ -6 \end{pmatrix}$.

$$\therefore a = -4, \quad b = -6.$$

21 c)

$$\frac{(x-2)^2}{4} + \frac{y^2}{9} = 1 \mapsto \frac{(x+2)^2}{4} + \frac{(y+6)^2}{9} = 1 \mapsto \frac{(-x+2)^2}{4} + \frac{(y+6)^2}{9} = 1.$$

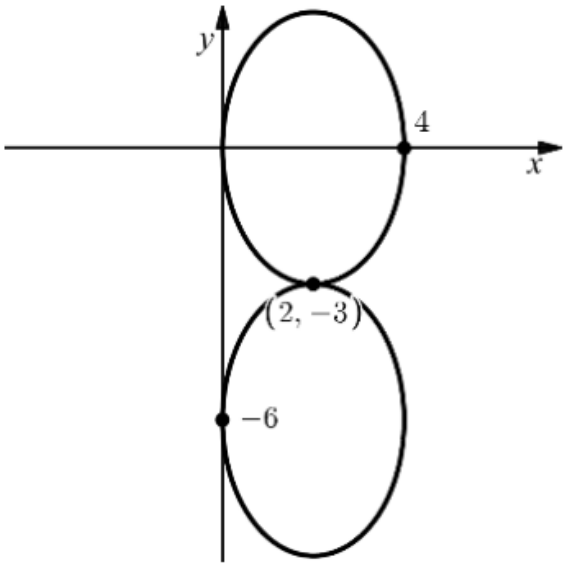
$$\therefore \frac{(x-2)^2}{4} + \frac{(y+6)^2}{9} = 1.$$

21 d)

$(2, -3)$.

21 e)

Sketch both C_1 and C_2 on the same axes.



21 f)

$x = 0$, $x = 4$ and $y = -3$.

22 a)

$(2, 0)$, $(0, \sqrt{2})$, $(-2, 0)$ and $(0, -\sqrt{2})$.

22 b)

Translation by vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$: $(x, y) \mapsto (x - k, y)$.

$$\frac{x^2}{4} + \frac{y^2}{2} = 1 \mapsto \frac{(x-k)^2}{4} + \frac{y^2}{2} = 1.$$

Enlargement of scale factor $\sqrt{3}$: $(x, y) \mapsto (\frac{x}{\sqrt{3}}, \frac{y}{\sqrt{3}})$.

$$\frac{(x-k)^2}{4} + \frac{y^2}{2} = 1 \mapsto \frac{(x/\sqrt{3}-k)^2}{4} + \frac{(y/\sqrt{3})^2}{2} = 1.$$

$$\therefore \frac{(x-\sqrt{3}k)^2}{12} + \frac{y^2}{6} = 1.$$

22 c)

When $y = -\frac{1}{2}x + 5$:

$$\frac{(x - \sqrt{3}k)^2}{12} + \frac{(-\frac{1}{2}x + 5)^2}{6} = 1 \Rightarrow (x - \sqrt{3}k)^2 + 2(-\frac{1}{2}x + 5)^2 = 12.$$

$$\Rightarrow x^2 - 2\sqrt{3}kx + 3k^2 + \frac{1}{2}x^2 - 10x + 50 = 12 \Rightarrow 2x^2 - 4\sqrt{3}kx + 6k^2 + x^2 - 20x + 76 = 0.$$

$$\therefore 3x^2 - (4\sqrt{3}k + 20)x + 6k^2 + 76 = 0.$$

22 d)

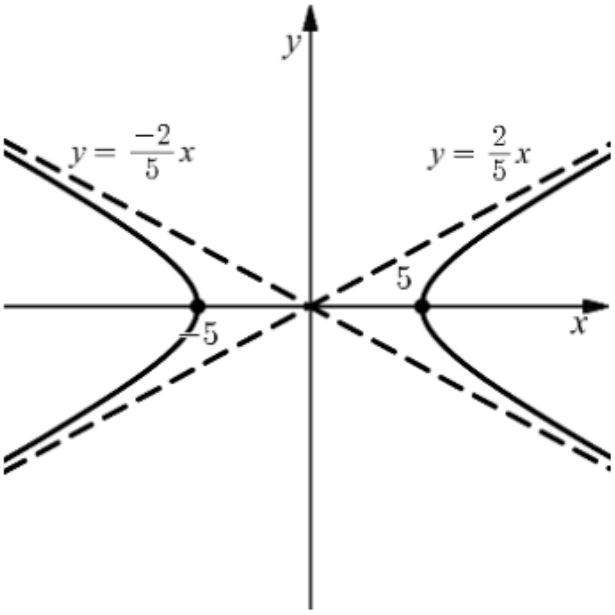
One repeated solution requires $b^2 - 4ac = 0$:

$$(4\sqrt{3}k + 20)^2 - 4(3)(6k^2 + 76) = 0 \Rightarrow 48k^2 + 160\sqrt{3}k + 400 - 72k^2 - 912 = 0.$$

$$\Rightarrow 24k^2 - 160\sqrt{3}k + 512 = 0 \Rightarrow k = \frac{4\sqrt{3}}{3} \text{ or } \frac{16\sqrt{3}}{3}.$$

23 a)

Sketch the hyperbola H.



23 b)

$$\frac{x^2}{25} - \frac{(x + k)^2}{4} = 1 \Rightarrow 4x^2 - 25(x + k)^2 = 100 \Rightarrow 4x^2 - 25x^2 - 50kx - 25k^2 = 100.$$

$$\Rightarrow 21x^2 + 50kx + 100 + 25k^2 = 0.$$

Two distinct solutions require $b^2 - 4ac > 0$:

$$(50k)^2 - 4(21)(100 + 25k^2) > 0 \Rightarrow 2500k^2 - 2100k^2 - 8400 > 0 \Rightarrow 400k^2 - 8400 > 0.$$

$$\Rightarrow k^2 - 21 > 0 \Rightarrow k > \sqrt{21} \text{ and } k < -\sqrt{21}.$$

23 c)

$$\frac{(x-a)^2}{25} - \frac{(y-b)^2}{4} = 1 \Rightarrow 4(x-a)^2 - 25(y-b)^2 = 100.$$
$$\Rightarrow 4x^2 - 8ax + 4a^2 - 25y^2 + 50by - 25b^2 - 100 = 0.$$

So $-8a = -32 \Rightarrow a = 4$ and $50b = 50 \Rightarrow b = 1$. Substitute:

$$4x^2 - 32x + 64 - 25y^2 + 50y - 25 - 100 = c \Rightarrow c = 61.$$
$$\therefore a = 4, \quad b = 1, \quad c = 61.$$

23 d)

Let $y = x + p$ be a line parallel to $y = x$.

$$4x^2 - 25(x+p)^2 - 32x + 50(x+p) = 61 \Rightarrow 21x^2 + (50p - 18)x + 25p^2 - 50p + 61 = 0.$$

A tangent requires $b^2 - 4ac = 0$:

$$(50p-18)^2 - 4(21)(25p^2 - 50p + 61) = 0 \Rightarrow 400p^2 + 2400p - 4800 = 0.$$
$$\therefore p = -3 \pm \sqrt{21}.$$

Thus $y = x - 3 + \sqrt{21}$ and $y = x - 3 - \sqrt{21}$.

24 a)

$$\frac{x^2}{4} + \frac{y^2}{100} = 1 \Rightarrow 25x^2 + (2x+k)^2 = 100 \Rightarrow 29x^2 + 4kx + k^2 - 100 = 0.$$

Two distinct solutions require $b^2 - 4ac > 0$:

$$(4k)^2 - 4(29)(k^2 - 100) > 0 \Rightarrow 16k^2 - 116k^2 + 11600 > 0 \Rightarrow -100k^2 + 11600 > 0.$$
$$\Rightarrow k^2 - 116 < 0 \Rightarrow -\sqrt{116} < k < \sqrt{116}.$$

24 b)

$$\frac{(x-a)^2}{4} + \frac{(y-b)^2}{100} = 1 \Rightarrow 25(x-a)^2 + (y-b)^2 = 100.$$
$$\Rightarrow 25x^2 - 50ax + 25a^2 + y^2 - 2by + b^2 - 100 = 0.$$

So $-50a = -150 \Rightarrow a = 3$ and $-2b = 10 \Rightarrow b = -5$. Substitute:

$$25x^2 - 150x + 225 + y^2 + 10y + 25 - 100 = c \Rightarrow c = -150.$$
$$\therefore a = 3, \quad b = -5, \quad c = -150.$$

24 c)

Let $y = 2x + p$ be a line parallel to $y = 2x$.

$$25x^2 + (2x+p)^2 - 150x + 10(2x+p) = -150 \Rightarrow 29x^2 + (4p - 130)x + p^2 + 10p + 150 = 0.$$

A tangent requires $b^2 - 4ac = 0$:

$$(4p-130)^2 - 4(29)(p^2 + 10p + 150) = 0 \Rightarrow p^2 + 22p + 5 = 0.$$
$$\therefore p = -11 \pm 2\sqrt{29}.$$

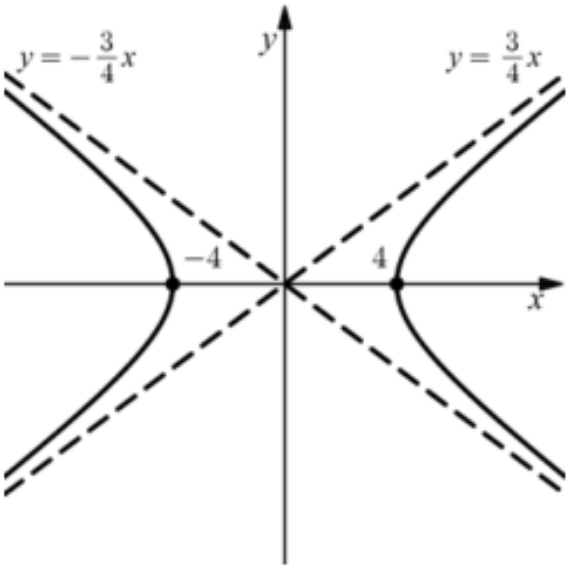
Thus $y = 2x - 11 + 2\sqrt{29}$ and $y = 2x - 11 - 2\sqrt{29}$.

25 a)

Hyperbola.

25 b)

Sketch the curve C_1.



25 c)

The asymptotes $y = 3x - k$ and $y = k - 3x$ intersect at the centre of C_2:

$$3x - k = k - 3x \Rightarrow 6x = 2k \Rightarrow x = \frac{k}{3}.$$

When $x = \frac{k}{3}$, $y = 3 \times \frac{k}{3} - k = 0$, so the centre is $\left(\frac{k}{3}, 0\right)$. When $k = 6$, the asymptotes of C_2 pass through $(2, 0)$, so the translation is by vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

$$\frac{x^2}{16} - \frac{y^2}{9} = 1 \mapsto \frac{(x-2)^2}{16} - \frac{y^2}{9} = 1.$$

Consider the original asymptote $y = \frac{3}{4}x$:

$$y = \frac{3}{4}x \mapsto y = \frac{3}{4}(x-2) = \frac{3}{4}x - \frac{6}{4}.$$

So the stretch is a stretch parallel to the y-axis, scale factor 4:

$$\frac{(x-2)^2}{16} - \frac{(y/4)^2}{9} = 1 \mapsto \frac{(x-2)^2}{16} - \frac{y^2}{144} = 1.$$

$$\therefore \frac{(x-2)^2}{16} - \frac{y^2}{144} = 1.$$

25 d)

As k is positive, it only gives a translation in the positive x-direction by vector $\begin{pmatrix} \frac{k}{3} \\ 0 \end{pmatrix}$:

$$\frac{(x - k/3)^2}{16} - \frac{y^2}{144} = 1.$$

Three intersections occur where the curves are tangent at one point. When $y = 0$:

$$\frac{x^2}{16} = 1 \Rightarrow x = \pm 4, \qquad \frac{(x - k/3)^2}{16} = 1 \Rightarrow x = \frac{k}{3} \pm 4.$$

So $4 = \frac{k}{3} - 4$ and $k = 24$.